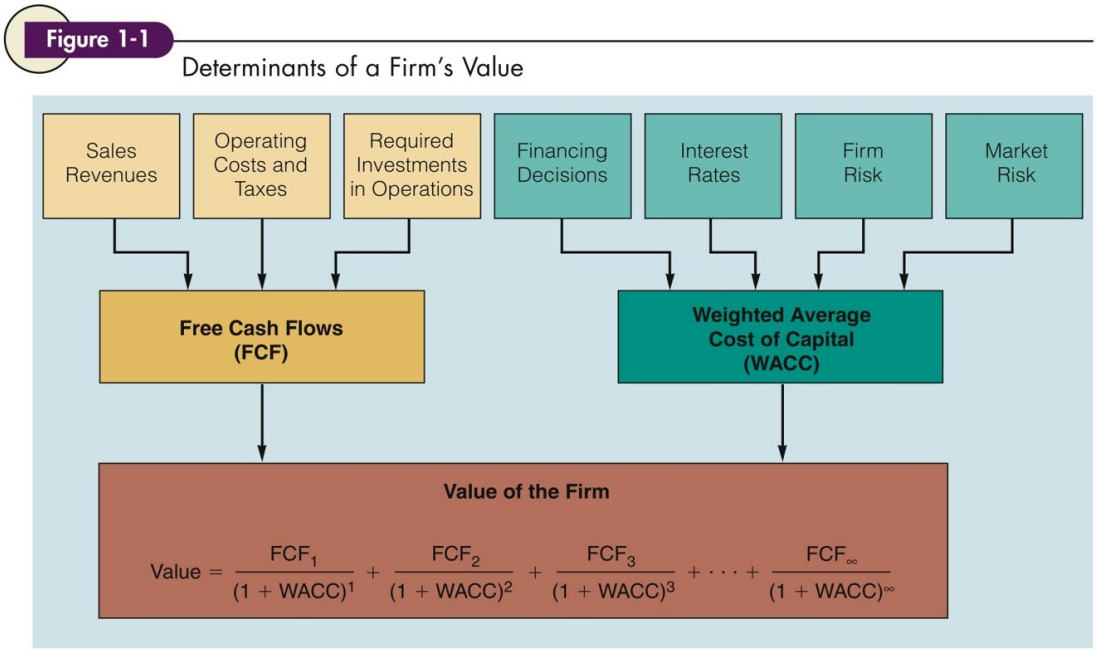
FMGT 3510 – Finance 1
Fall 2026

Chapter 2

Financial Statements, Taxes, and Cash Flows

How can we use financial statements to derive Free Cash Flows (FCF)?

To do so, we need to review accounting standards and the main financial statements.



Accounting standards and financial statements

Companies report their financial results by producing financial statements:

- Statement of Financial Position: the "Balance Sheet"
- Statement of Profit and Loss and Other Comprehensive Income: the "Income Statement"
- Statement of Changes in Equity
- Statement of Cash Flows.

In Canada, these statements are prepared in accordance with **International Financial Reporting Standards (IFRS)**

However privately-held companies can opt to use a simpler standard: **Accounting Standards for Private Enterprise (ASPE)**

Why would a private firm (i.e. not listed) opt for IFRS?

Balance Sheet – review (1of 3)

ACME Limited Balance Sheet as at December, 31st

in millions of $	2023	2022		2023	2022
Assets			*Liabilities and Equity*		
Cash	**5.0**	5.0	Accounts payable	28.2	24.5
Short-term investments	18.2	15.5	Short term bank loans	**16.7**	15.5
Accounts receivable	**16.5**	13.2	Accruals	2.0	4.0
Inventories	**15.3**	14.3	Total current liabilities	46.9	44.0
Total current assets	55.0	48.0			
			Long-term debt	**70.2**	63.7
Fixed assets	110.8	98.4	Shareholders' equity		
Less: accumulated depreciation	-18.7	-17.5	Common stock	**5.0**	5.0
Net fixed assets	**92.1**	80.9	Retained earnings	**25.0**	16.2
			Total equity	**30.0**	21.2
Total assets	147.1	128.9	Total liabilities & equity	147.1	128.9

Balance Sheet (BS) – review (2of 3)

The balance sheet is a snapshot at one moment in time of what a firm owns (assets) and what a firm owes (liabilities).

B**alance sheet equation**: **ASSETS – LIABILITIES = EQUITY**

Shareholders' equity reflects a residual interest in the assets of the firm (i.e. the claims of shareholders are subordinate to the claims of debt holders).

Assets are listed from top to bottom in order of liquidity:

Cash and equivalents
Short term investments
Receivables
Inventory
Long term assets

Current Assets < 1 year

Balance Sheet (BS) – review (3 of 3)

Liabilities are listed in order of how soon they must be paid:

- ***Payables***
- ***Notes payables***
- ***Accruals***

} Current Liabilities < 1 year

- Long term liabilities

Equity consists of

- **Preferred stock** (if any)
- **Common stock** and **retained earnings**

Remember - Main issues with the balance sheet:

- Assets are normally recorded at **historic cost ; the numbers shown in the BS do not reflect current market value**.
- Many of a firm's **most valuable assets**, some **intangible assets** such as good management, good reputation, employees' talent and dedication, **do not appear on the balance sheet at all.**

Income Statement – review (1 of 3)

ACME ltd. Income statement for the years		
in million of $	2023	2022
Sales	205.0	176.1
COGS	159.2	147.3
Other expenses	23.1	20.6
EBITDA	22.7	8.2
Depreciation	1.2	1.1
EBIT	21.5	7.1
Interest expense	6.0	4.6
EBT	15.5	2.5
Taxes	3.7	0.6
Net Income	11.8	1.9
Dividends	3.0	
Tax rate	24%	24%

Income Statement – review (2 of 3)

The income statement measures performance over a period of time (normally a year).

Income statement equation: Net Income = Revenues less Expenses.

IFRS also allows for the inclusion of **Other Comprehensive Income** ("OCI").
OCI account records **unrealized gains and losses** on assets such as investments, derivatives or foreign exchange hedges. These items are set out in a separate statement and recorded separately in shareholders' equity.

The Acme Ltd. financials do not show any OCI.

Income Statement – review (3 of 3)

There is a **disconnect between** the **net income** on the income statement **and the cash flow:**

The income statement is prepared on an **accrual basis** to accommodate the **matching principle** (i.e. match revenues with the expenses associated with producing them).

The **timing** of recorded revenues and expenses may not correspond with the timing of cash receipts and payments.

There are also **non-cash items** that appear on the income statement, such as:

- Depreciation/amortization
- Bad debt expense
- Options expense
- Gains/losses on disposition of assets

Measures of profitability

Operating profitability: profitability of the business' assets excluding the impact of non-operating expenses such as interest and taxes.

Operating profitability is often measured using the **E**arnings **B**efore **I**nterest and **T**axes (**EBIT**), also called **Operating Income.**

What is Acme's 2023 EBIT? $21.5

Another common operating profitability measure is the **E**arnings **B**efore **I**nterest **T**axes **D**epreciation and **A**mortization (**EBITDA**).

$$EBITDA = Sales - Operating\ Costs = EBIT + Depreciation\ and\ Amortization$$

Its calculation adds back a major non-cash expense (depreciation) in addition to adding back interest and taxes.

What is Acme Ltd.'s 2023 EBITDA? **EBITDA = EBIT + DA = $21.5 + $1.2 = $22.7**

Statement of Cash Flows – review (1 of 2)

The statement of cash flows shows (SCF) how the firm's **cash position** changed during the period the question.

In this class we will not review how to construct the SCF. We will focus on interpreting the numbers.

What does Acme's 2023 Cash From Operating Activities tell us?

= $10.4 *Day-to-day business operations are generating cash.*

Cash From Investing Activities?

= ($15.1) *Cash has been used to acquire more long-term assets and short-term investments (STI).*

Cash From Financing Activities?

= $4.7 *Some capital was needed to offset the cash used in investing activities. Overall, the cash balance did not change compared to 2022.*

Which one is the most important? CF from operations, investing or financing activities?

Statement of Cash Flows – review (2 of 2)

ACME Ltd. Statement of Cash Flows for the year 2023				
Operating Activities				
	Net Income		11.8	
Adjustments				
	add back depreciation (non cash expense)		1.2	
	due to changes in working capital			*Increase*
		Change in AR	-3.3	*Increase*
		Change in inventories	-1.0	*Increase*
		Change in AP	3.7	*Increase*
		Change in Accruals	-2.0	*Decrease*
Net Cash from Operating Activities			10.4	
Investing Activities				
	Cash used to acquire new fixed assets		-12.4	
	Purchase of short-term investments		-2.7	
Net Cash used in Investing Activities			-15.1	
Financing Activities				
	Change in Notes Payable		1.2	*Increase*
	Change in Long-term debt		6.5	*Increase*
	Dividends payment		-3.0	
Net Cash provided by Financing Activities			**4.7**	
Net Change in cash			0.0	
	Cash balance at the beginning of 2023		5.0	
	Cash balance at the end of 2023		5.0	

10.4 -15.1 + 4.7 = 0

Balance Sheet cash balance at the end of 2023

Net Cash Flows

Net cash flow (**NCF**) is an alternative measure of profitability which adjusts net income for the fact that some expenses, principally depreciation and amortization, do not involve the use of cash

$$\text{Net cash flow} = \text{Net income available to common shareholders} - \text{Noncash revenues} + \text{Noncash charges} \quad (2\text{-}1)$$

More often it is more simply calculated by analysts as **net income plus depreciation and amortization:**

$$\text{Net cash flow} = \text{Net income available to common shareholders} + \text{Depreciation and amortization} \quad (2\text{-}2)$$

Warning: NCF and Free Cash Flow (FCF) are two different concepts (*see next slides for FCF calculation*). NCF is a widely-used rough estimate for **profit the firm would earn if it did not replace its fixed assets as they wear out** (*i.e. if no depreciation/amortization expense*).

2023 NCF for Acme is $11.8 mill + $1.2 mill = $13.0 mill.

Free Cash Flows (FCF) Modifying accounting data for use in managerial decision making (1 of 2)

At this point we will **re-organize** Acme's financial statements to make them more useful for decision making.

We will divide Acme's **assets** into **two categories**:

- **Operating Assets:** assets needed to run the business
- **Non-Operating Assets:** assets considered as "surplus" or "excess assets".

We will also divide Acme's **liabilities** into two categories:

- **Investor Supplied Capital:** represent funding from investors
- **Operating Current Liabilities:** do not represent funding from investors.

  Investors (lenders and shareholders) require a return on investment.

We will further divide Acme's assets into **Net Operating Working Capital** and **Operating Long Term Assets**

Once we have re-organized the financial statements, we will be able to calculate the amount of **Free Cash Flow** produced by Acme.

Free Cash Flows (FCF) Modifying accounting data for use in managerial decision making (2 of 2)

Why is this important?

The best way to estimate the **value of a firm's assets** is by calculating the **present value of its future Free Cash Flows.**

- Free cash flow is the foundation of **corporate valuation**.
- Greater long-term future free cash flows, all other things being equal, means greater business value.
- Estimating a business' value based on free cash flow, allows us **to see if management's decisions are in fact increasing shareholders' wealth.**

Free Cash Flows (FCF)

Operating and non-operating current assets (1 of 2)

The starting point in calculating free cash flow is correctly categorizing our firm's assets and liabilities.

Operating Current Assets 2023: Current assets needed to operate the business. We will treat short term investments as "excess assets" that are not required to operate Acme.

Operating Current Assets = **$5.0 + $16.5 + $15.3 = $36.8**

Cash + AR + INV

Non-operating Current Assets 2023 = Short term investments = **$18.2**

Free Cash Flows (FCF)

Operating and non-operating current assets (2 of 2)

ACME Limited Balance Sheet as at December, 31st

in millions of $	2023	2022		2023	2022
Assets			*Liabilities and Equity*		
Cash	**5.0**	5.0	Accounts payable	28.2	24.5
Short-term investments	18.2	15.5	Short term bank loans	**16.7**	15.5
Accounts receivable	**16.5**	13.2	Accruals	2.0	4.0
Inventories	**15.3**	14.3	Total current liabilities	46.9	44.0
Total current assets	55.0	48.0			
			Long-term debt	**70.2**	63.7
Fixed assets	110.8	98.4	Shareholders' equity		
Less: accumulated depreciation	-18.7	-17.5	Common stock	**5.0**	5.0
Net fixed assets	**92.1**	80.9	Retained earnings	**25.0**	16.2
			Total equity	**30.0**	21.2
Total assets	147.1	128.9	Total liabilities & equity	147.1	128.9

	2023	
Operating Current Assets		
Cash	$	5.0
AR	$	16.5
Inventories	$	15.3
	$	36.8

Free Cash Flows (FCF) - Operating current Liabilities, NOWC, Investors supplied capital, TNOC (1 of 4)

Operating Current Liabilities = Accounts payable, accruals.

*These liabilities are often called **"spontaneous liabilities"** because they increase roughly automatically with sales.*

Investor Supplied Capital = Short-term and long-term interest paying debt, and equity.

*The providers of this capital **require a rate of return on their investment**.*

Net Operating Working Capital (NOWC)

= Current Operating Assets - Operating Current Liabilities

Example – Acme 2023

Operating Current Liabilities = 28.2 + $2.0 = AP + accruals = $30.2

Investor supplied capital = $16.7 + 70.2 + 30.0 = ST.D + LT.D + Equity = $116.9

NOWC = $36.8 - $30.2 = $6.6

Most of the current operating assets are financed by operating current liabilities but $6.6 are not and must be financed by investors.

Free Cash Flows (FCF) - Operating current Liabilities, NOWC, Investors supplied capital, TNOC (2 of 4)

ACME Limited Balance Sheet as at December, 31st

in millions of $	2023	2022		2023	2022
Assets			*Liabilities and Equity*		
Cash	**5.0**	5.0	Accounts payable	28.2	24.5
Short-term investments	18.2	15.5	Short term bank loans	**16.7**	15.5
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Net fixed assets	**92.1**	80.9	Retained earnings	**25.0**	16.2
			Total equity	**30.0**	21.2
Total assets	147.1	128.9	Total liabilities & equity	147.1	128.9

	2023	
Operating Current Assets		
Cash	$	5.0
AR	$	16.5
Inventories	$	15.3
	$	36.8
Less:		
Operating Current Liabilities		
AP	$	28.2
Accruals	$	2.0
	$	30.2
Net Operating Working Capital	$	6.6

Free Cash Flows (FCF) - Operating current Liabilities, NOWC, Investors supplied capital, TNOC (3 of 4)

Total Net Operating Capital (TNOC)
= Net Operating Working Capital + Operating Long Term Assets

Example – Acme 2023 – Based on our previous calculations and balance sheet data:
Acme has **$92.1** of long-term assets, net of depreciation
TNOC = $6.6 + $92.1 = $98.7

$98.7 *of total net operating capital is used in Acme's business operations.*
Acme requires ***$98.7*** *of investor capital to finance these operating assets.*

Free Cash Flows (FCF) - Operating current Liabilities, NOWC, Investors supplied capital, TNOC (4 of 4)

ACME Limited Balance Sheet as at December, 31st

in millions of $	2023	2022
Assets		
Cash	5.0	5.0
Short-term investments	18.2	15.5
Accounts receivable	16.5	13.2
Inventories	15.3	14.3
Total current assets	55.0	48.0
Fixed assets	110.8	98.4
Less: accumulated depreciation	-18.7	-17.5
Net fixed assets	92.1	80.9
Total assets	147.1	128.9

	2023	2022
Liabilities and Equity		
Accounts payable	28.2	24.5
Short term bank loans	16.7	15.5
Accruals	2.0	4.0
Total current liabilities	46.9	44.0
Long-term debt	70.2	63.7
Shareholders' equity		
Common stock	5.0	5.0
Retained earnings	25.0	16.2
Total equity	30.0	21.2
Total liabilities & equity	147.1	128.9

	2023	
Operating Current Assets		
Cash	$	5.0
AR	$	16.5
Inventories	$	15.3
	$	36.8
Less:		
Operating Current Liabilities		
AP	$	28.2
Accruals	$	2.0
	$	30.2
Net Operating Working Capital	$	6.6
Net Operating Long-term Assets	$	92.1
Total Net Operating Capital	$	98.7

Free Cash Flows (FCF) calculation (1 of 3)

We now know how to calculate the amount of **total assets the business needs to support its operations**. **How much free cash flow do these assets produce and what can it be used for?**

Free Cash Flow: Cash flow available, that can be paid out to investors (*debtholders and shareholders*) **after the company has made all investments in fixed assets and working capital needed to sustain ongoing operations.**

- There are other definitions of free cash flow. You should always check the definition before using any data derived from a firm's financial statements.
- In this class we will calculate the **FCF available to both debtholders and shareholders**. Some analysts use the ***FCF to equity*** (available to shareholders only).

A **mature**, steadily growing business will likely have **positive free cash flow** year after year.

However free cash flow **may not always be a positive** amount, especially if a **business** is **growing rapidly.**

Free Cash Flows (FCF) calculation (2 of 3)

Personal example of free cash flow

You might have **after-tax income of $60,000** but **still have negative free cash flow** for the year if you also spend:

- $40,000 on a new car and
- $400,000 on a condo apartment.

In that case **negative free cash flow of $380,000 would have to be financed through loans and mortgages**.

After-tax earnings	$	60,000
Car Purchase	$	(40,000)
Condo Purchase	$	(400,000)
Free Cash Flow	$	(380,000)

Free Cash Flows (FCF) calculation (3 of 3)

What is true of individuals is also true of businesses, and vice versa.

Negative FCF today is not necessarily a bad thing if, in the case of a business, the **new assets acquired lead to a significant increase in future FCF**.

In the case of individuals, negative FCF as a result of buying a house can make sense. We no longer have to pay rent out of our after-tax earnings and therefore will increase our future FCF.

Our task now is to calculate Acme's free cash flow using the following method;

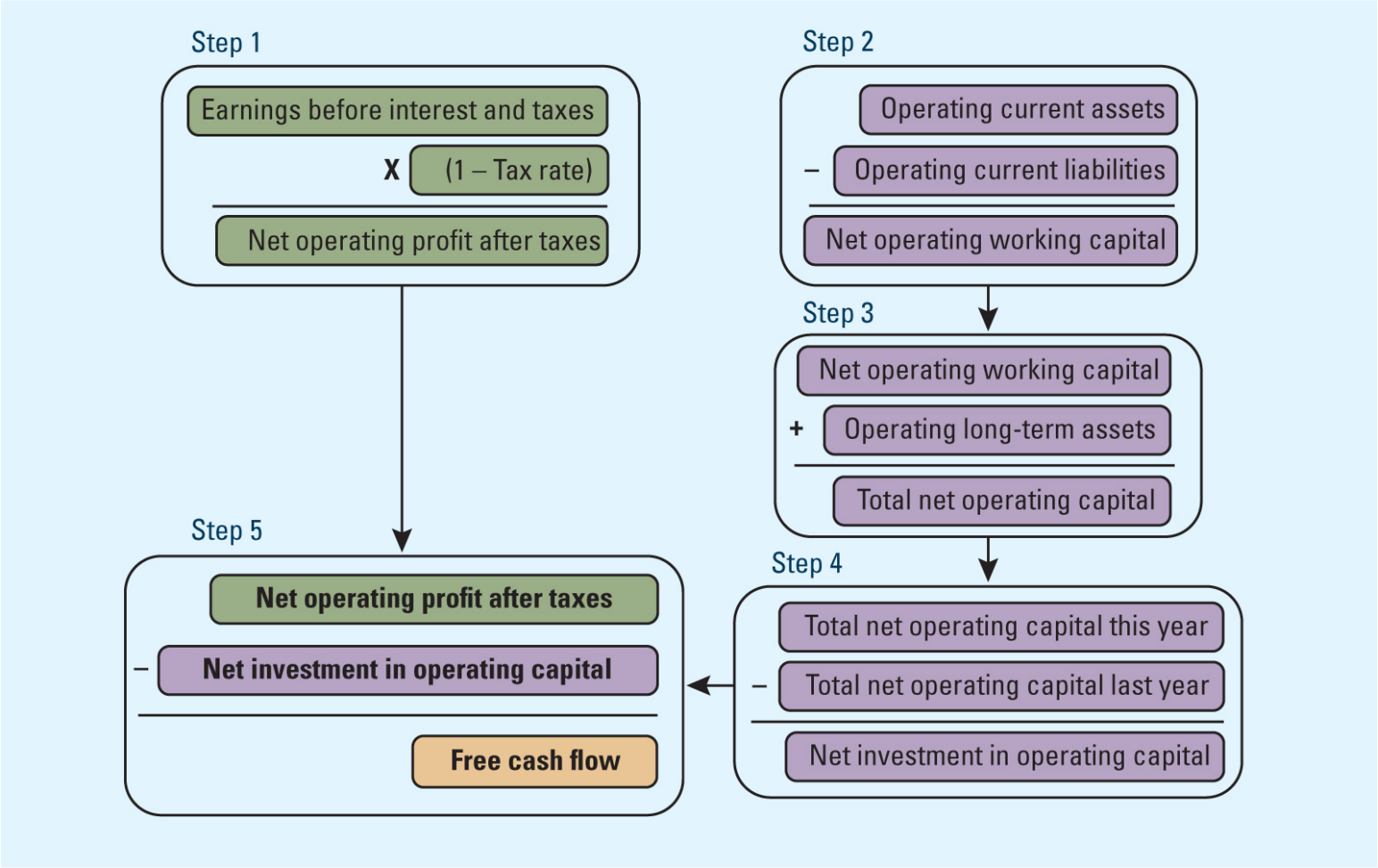
Net Operating Profit After-Tax ("NOPAT") = EBIT (1-T)

Less Additions to Net Operating Working Capital

Less Net additions to Operating Long Term Assets

Next slide: diagram summarizing Free Cash Flow calculation process.

FIGURE 2-1 Calculating Free Cash Flow



Acme's Free Cash Flows (FCF) calculation (1 of 5)

Step 1 Calculate NOPAT: $Net\ Operating\ Profit\ After\ Tax = EBIT(1 - tax\ rate)$

For Acme 2023 NOPAT = **$21.5 (1 - 0.24) =$16.34**

Steps 2 & 3 Calculate 2023 TNOC

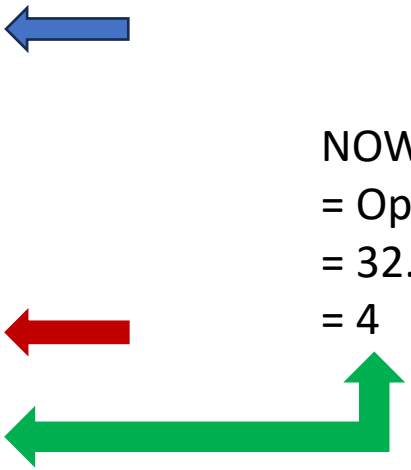
(we've done this earlier)

	2023	
Operating Current Assets		
Cash	$	5.0
AR	$	16.5
Inventories	$	15.3
	$	36.8
Less:		
Operating Current Liabilities		
AP	$	28.2
Accruals	$	2.0
	$	30.2
Net Operating Working Capital	$	6.6
Net Operating Long-term Assets	$	92.1
Total Net Operating Capital	$	98.7

Acme's Free Cash Flows (FCF) calculation (2 of 5)

Step 4a Calculate 2022 TNOC. Use the same method as for 2023

	2022	
Operating Current Assets		
Cash	$	5.0
AR	$	13.2
Inventories	$	14.3
	$	32.5
Less:		
Operating Current Liabilities		
AP	$	24.5
Accruals	$	4.0
	$	28.5
Net Operating Working Capital	$	4.0
Net Operating Long-term Assets	$	80.9
Total Net Operating Capital	$	84.9



NOWC (end of 2022)
= OpCA – OpCLiab
= 32.5 - 28.5
= 4

Acme's Free Cash Flows (FCF) calculation (3 of 5)

<u>Step 4b</u> Calculate the Net Investment in TNOC for 2023: the amount by which TNOC changed from 2022 to 2023.

	2023	2022
Operating Current Assets		
Cash	$ 5.0	$ 5.0
AR	$ 16.5	$ 13.2
Inventories	$ 15.3	$ 14.3
	$ 36.8	$ 32.5
Less:		
Operating Current Liabilities		
AP	$ 28.2	$ 24.5
Accruals	$ 2.0	$ 4.0
	$ 30.2	$ 28.5
Net Operating Working Capital	$ 6.6	$ 4.0
Net Operating Long-term Assets	$ 92.1	$ 80.9
Total Net Operating Capital	$ 98.7	$ 84.9
	$ 84.9	
Net Investment in TNOC in 2023	**$ 13.8**	

98.7 – 84.9

Acme's Free Cash Flows (FCF) calculation (4 of 5)

<u>Step 5</u> Calculate Free Cash Flow (FCF).

	2023	
EBIT	$	21.5
Taxes	$	5.2
NOPAT	$	16.3
Net Investment in Operating Capital	$	(13.8)
Free Cash Flow	**$**	**2.5**

Acme's **free cash flow in 2023 was $2.5 million.**

2023's **net income of $11.8 million** clearly does not measure the amount of cash produced by Acme's business operations.

The free cash flow of **$2.5 million** shows how much **cash was available to be distributed to its investors (lenders and shareholders).**

This amount is exactly what lenders and shareholders need to value Acme correctly.

Acme's Free Cash Flows (FCF) calculation (5 of 5)

Here's a reconciliation of how Acme's free cash flow was used in 2023

Financing of (use of) free cash flow			
Pay after-tax interest to debt holders	$	(4.6)	Interest adjusted for taxes: $6.0 x (1 - 0.24)
Change in debt principal	$	7.7	Bank loan increased $1.2; long-term debt increased $6.5
Pay dividends to shareholders	$	(3.0)	
Issue (repurchase) stock	$	-	
(Increase) decrease STI	$	(2.7)	STI increased $2.7
Equal to free cash flow	$	(2.5)	rounding

Examples – different companies use CF in different ways

2023 results in $ millions

	Ford	Meta	Rogers	Uber
Operations	$ 14,918	$ 71,113	$ 5,221	$ 3,585
Investing	$ (17,628)	$ (24,495)	$ (20,200)	$ (3,226)
Financing	$ 2,584	$ (19,500)	$ 2,479	$ (95)
Net	$ (126)	$ 27,118	$ (12,500)	$ 264

Measuring performance

Return on Invested Capital (ROIC): a measure of the profitability of the firm's assets:
$$\boldsymbol{ROIC = NOPAT / Total\ Net\ Operating\ Capital}$$

ACME's 2023 ROIC = **$16.3 / $98.7 = 16.5%**

If Acme's cost of capital is 10% did Acme create shareholder value in 2023?

Yes: 16.5% > 10.0%